1

Prepare

while the integrate-and-fire model is

$$\begin{cases} C_m \frac{dV_m(t)}{dt} = I(t) - g_{leak}(V_m(t) - V_{leak}) \\ V(t^-_{spike}) = V_{threshold} \\ V(t^+_{spike}) = V_{resting} \end{cases} \tag{2}$$

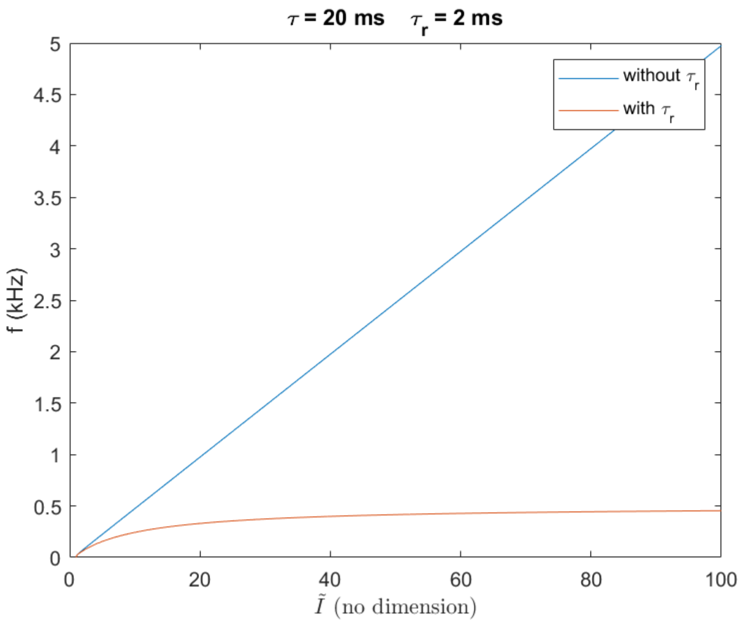
Let us define

$$\begin{cases} \tilde{V} := \frac{V - V_{resting}}{V_{threshold} - V_{resting}} \\ I_c := g_{leak}(V_{threshold} - V_{resting}) \\ \tau := RC \\ \tilde{I} := \frac{I_e}{I_c} \end{cases} \tag{3}$$

then we get the normalized form of the integrate-and-fire model

$$\begin{cases} \tau \frac{d\tilde{V}(t)}{dt} = \tilde{I}(t) - \tilde{V}(t) \\ \tilde{V}(t^-_{spike}) = 1 \\ \tilde{V}(t^+_{spike}) = 0 \end{cases} \tag{4}$$

a



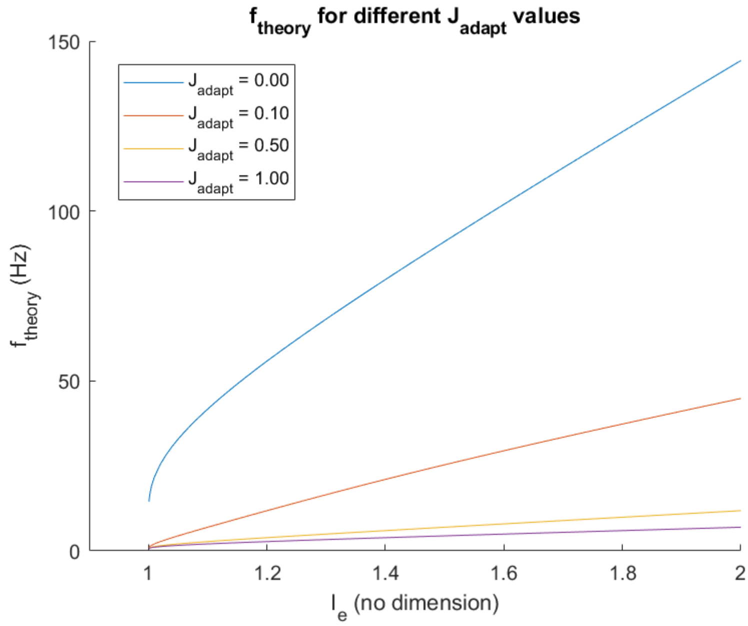
- Without τ_r:linear
- Wiht τ_r:

$$\lim_{\tilde{I}\to\infty} f = 500Hz$$

b

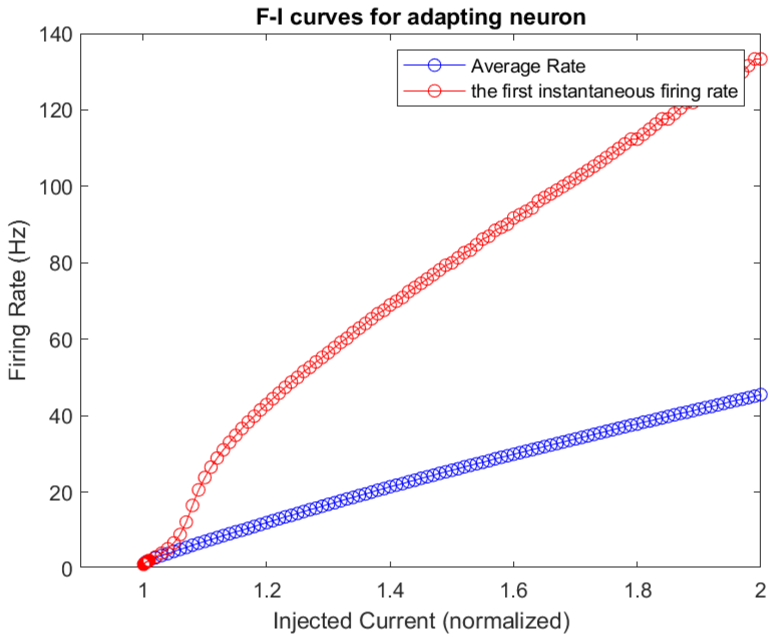
Analytic Solution of the Differential Equation

- the Algebraic Equation: see GoodNote
- Analytic Solution of the Algebraic Equation: can't get
- Numerical Solution of the Algebraic Equation: see below
  - conclusion: $J \nearrow, f \searrow$

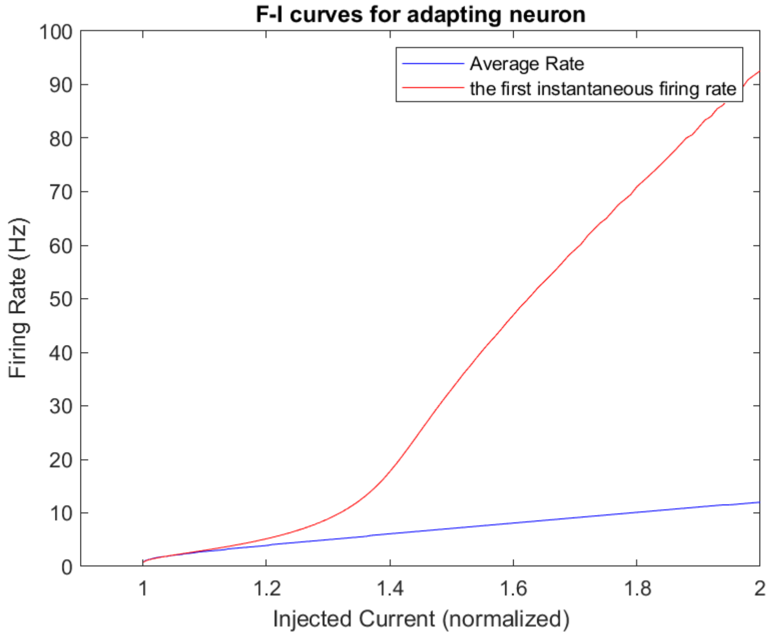


Numerical Solution of the Differential Equation

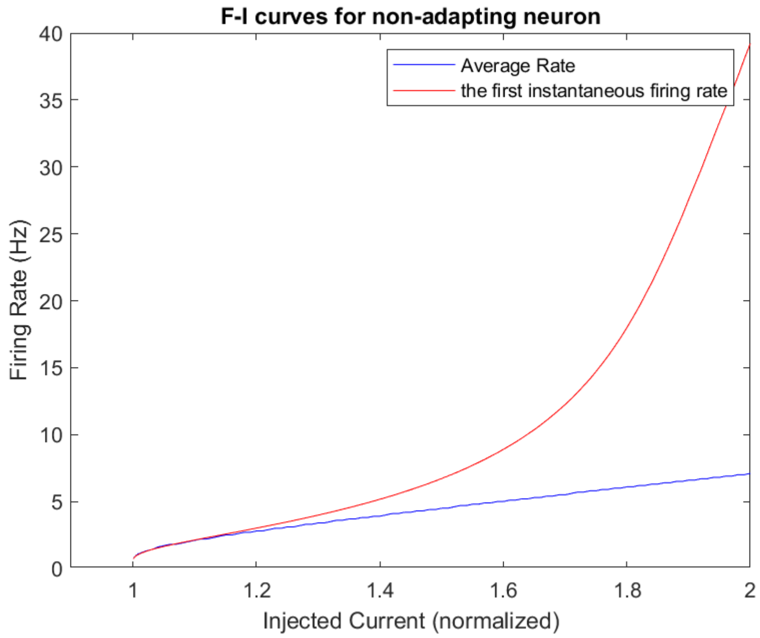
$J_{adapt} = 0.1$



$J_{adapt} = 0.5$

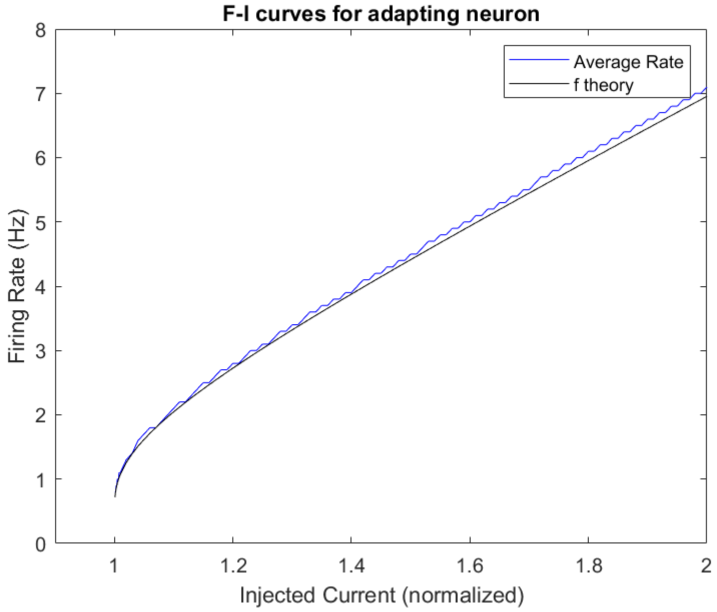


$J_{adapt} = 1.0$



Analytic vs Numerical solution of the ODE

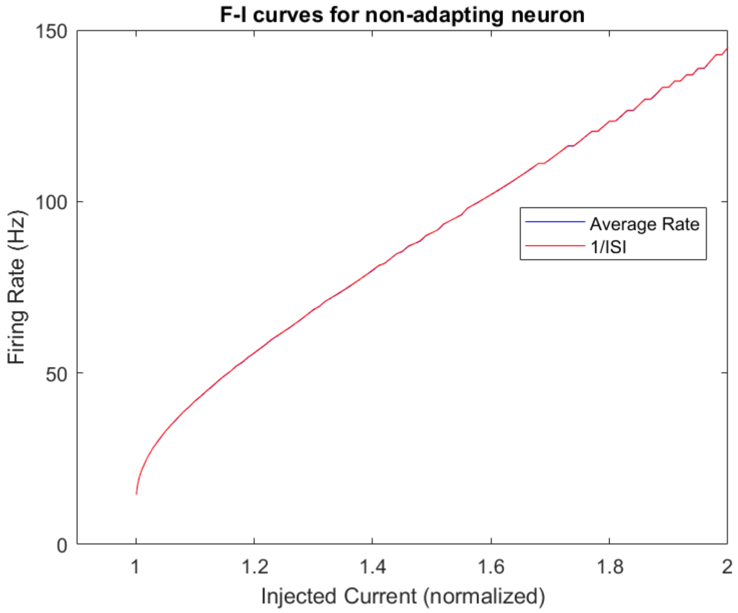
$J_{adapt} = 1.0$



- conclusion: very similar.

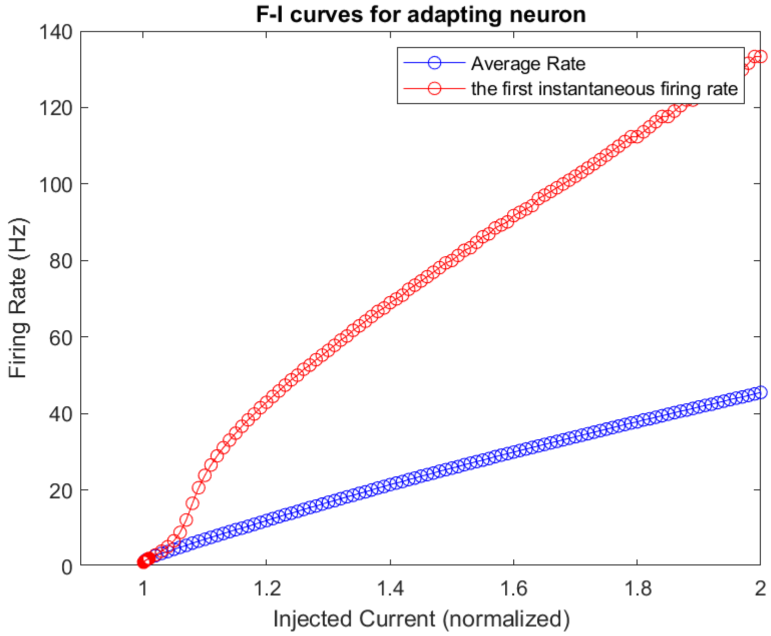
C

$J_{adapt} = 0$



- conclusion: very similar.

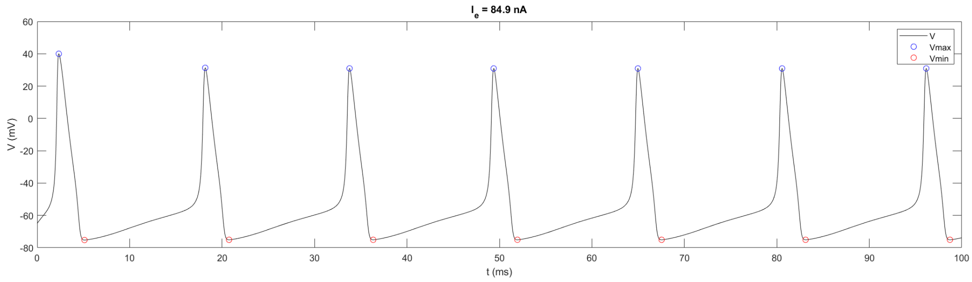
$J_{adapt} = 0.1$

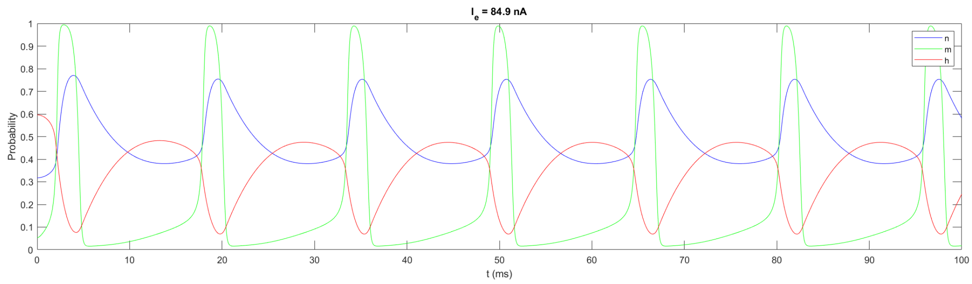


- conclusion: not similar.

2

a





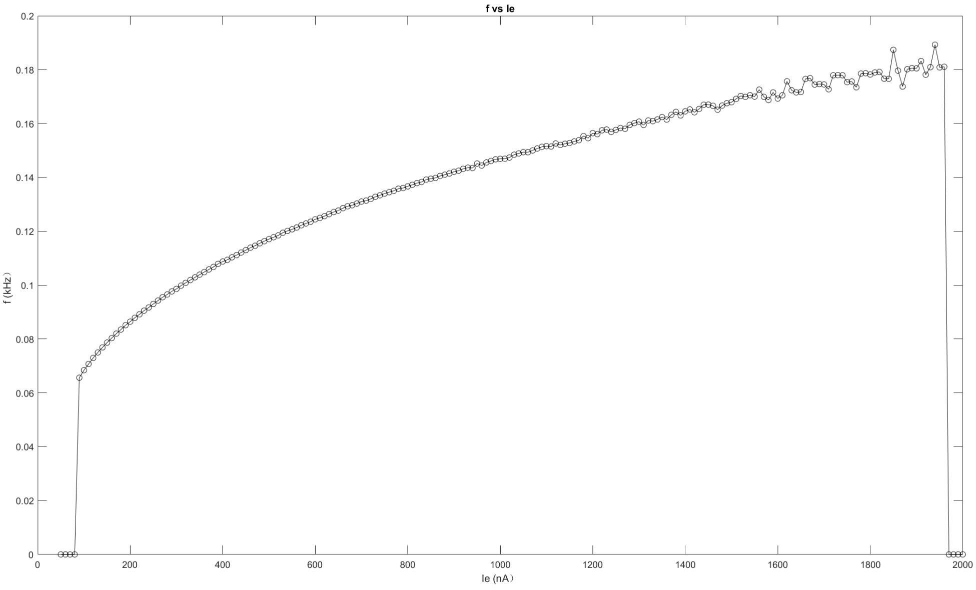
- conclusion: m is faster.

b

Threshold

My threshold is 84.5-84.6 nA.

$f \sim I_e$



conclusion: 0 -> log-like -> 0 (The fluctuation may due to my bad algorithm or the bad performance of ode45)